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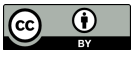
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A standardized approach to population equivalents for food consumption analysis and socio-metabolic research: insights from France

Barbara Redlingshöfer, Tanguy Fardet, Caroline Petit

Abstract

Food consumption is a significant driver of environmental pressure, particularly in urban areas. As for mobility and buildings, which are other major impact sources, different types of population are responsible for the associated demand (residents, workers, tourists, etc). Despite preliminary work, standardized methods to quantify the contribution of different populations to social metabolism remain limited.

This paper provides a comprehensive framework to estimate the populations present in an area and their food consumption, over a given period of time, based on broadly available census and statistical data. By using population equivalents, we provide a standardized and consistent approach to integrate the different types of population and their respective time spent in the area, as well as their share of daily food intake. A case study of France reveals that, in 85% of municipalities and 75% of inter-municipal cooperation areas, the present population is smaller than the resident population. The eating population is usually 5 to 20% smaller than the resident population, because children, teenagers, and commuters consume less food overall. In most metropolitan areas, the annual population increase due to commuters and tourists is offset by the residents' departures during holiday periods, bringing the present population quite close to the resident population. Only touristic areas attract additional populations to such an extent that the present population becomes substantially larger than the resident population over a year. Previous approaches tend to overestimate the biomass directly associated with urban food consumption. Our population-based framework can help reduce uncertainties about biomass use in cities and improve the comparison of material consumption between different areas.

Keywords: urban metabolism, food flows, material flow analysis, food planning population equivalent, biomass

1. Introduction

1.1. Background

On a global scale, Tilman and Clark (2014) highlight the complex interconnections between urban population growth, the resulting nutritional transitions, and the carbon footprint of agricultural systems. Urban diets are often characterized by higher meat, sugar and fat content, leading to increased emissions from the agricultural systems and public health issues. A recurring question also concerns the capacity of agriculture to feed a growing population, a challenge shaped by multiple ecological, technological and socio-economic factors (Chatzimpiros and Harchaoui 2023). At the local scale, these concerns are for instance being translated into urban food policies that aim to drive food system transformation towards more sustainable practices and diets (Sonnino 2023).

Food consumption is a significant driver of environmental footprints, together with mobility and housing (Goldstein et al. 2017). These three sectors and their respective impacts involve different types of population (e.g. residents, tourists and commuters) in the area, their time of presence there, and their varying use patterns of material and energy during that time. However, these factors have not been taken into account in the impact analysis. In fact, the material and environmental footprint of food at territorial scale and the associated eating population have rarely been a focus of quantitative studies (Guibrunet and Sánchez Jiménez 2023), despite their potential role in environmental sustainability. Policymakers are nevertheless increasingly reengaging with food issues, notably to relocalize food production, shorten supply chains, reduce food waste and, more generally, alleviate food insecurity (Doernberg et al. 2019; IPES 2017; *Milan Urban Food Policy Pact*). This turn follows a long period of under-investment that began with the industrialization of food systems (Wiskerke 2009; Pothukuchi and Kaufman 2000).

Comprehensive quantitative assessments of food flows are necessary for efficient science-based local policies. Yet, there is a lack of conceptual frameworks to integrate the different types of population involved, how they use and inhabit a given area, and how much food they consume over time. Though this issue is not unfamiliar to some research fields, for instance population research or urban metabolism, as shown in the following review of the state-of-the-art, standardized and actionable methods to estimate the present population and its associated food consumption over time, at the territorial scale, whether urban or rural, are not yet available.

1.2. State-of-the-art in the estimation of population and food consumption

Concepts and methods to study the population present over time in a given area have been developed for decades in population research and statistics (Charles-Edwards et al. 2020; Charles-Edwards et al. 2021). A wide range of domains, such as emergency planning, tourism, transportation, and epidemiology, benefit from information about the present population in an area at a given time. As Panczak et al. (2020) show, the estimation of the temporary population, understood as the population driven by temporary mobility of residence without a change of usual residence at the level of cities and regions, can be traced back to the early 20th century. The origins of this field of study can be pinpointed to the 1950s in the United States. Smith (1989) later proposed a set of conceptual metrics (visitor-days and visitor years) to capture short-term mobility of a population and conceptualized the temporality of presence. This provided a standard measure to convert a population at a single point in time into a static annual stock equivalent – such as visitor-days or visitor-years – for comparison with resident populations. Smith also argued that researchers must select time criteria that specifically correspond to the resource or service being analyzed. As the focus of this study is food consumption, we use the year as the typical time for agricultural production and the consumption of the associated goods. However, we also consider other timescales (notably the touristic high season during summer) in specific areas.

Swanson and Tayman (2011) developed a detailed composition of the *de facto* population (as opposed to the *de jure* population) and its specific pattern of mobility. *De facto* population counts all individuals physically present in an area at a specific reference time (e.g. census night), regardless of their usual or legal residence. It thus includes several non-permanent populations in addition to the residents: visitors, homeless people, amenity-seeking and migrant seasonal populations, and non-resident daytime populations, present for only part of the day (González et al. 2008; Deville et al. 2014).

The increasing mobility and access to new data sources, particularly ICT-based data (information and communication technology, e.g. mobile phones, credit cards and Wi-Fi connections), exerts a significant impact on the concepts and methods used to capture the temporary population (Lévy et al. 2023). The constant refinement of the use of open-access census data and surveys to ensure methodological transparency and reproducibility for public policy is another trend relevant to this research (Vallée and Lenormand 2024). Although these considerations are deeply rooted in population studies, they also meet the needs of research areas that rely on harmonized per-capita indicators, such

as socio-metabolic research and environmental impact assessments.

The larger goal of this study is a quantitative assessment of the total food consumption of a population present in an urban area over a given time period. The scientific literature provides many examples of how food consumption can be estimated in infra-national areas such as cities and regions. Socio-metabolic studies commonly use material flow analysis (MFA) to analyze food flows related to a given area. To estimate one of these flows, food consumption, at the scale of cities or any given area, two different approaches can be distinguished (Goldstein et al. 2017):

Balance approaches: these apply an accounting method developed for a nation's economy-wide material flows (European Commission. Statistical Office of the European Union. 2018), further adjusted to evaluate flows at the scale of cities (Barles 2009); they conceptualize consumption as the amount of material in the domestic economy that remains within a specific area and does not leave it as export or waste, termed "domestic material consumption" (DMC)

Population-based approaches: these combine the population located in the area with per capita food availability or consumption to assess the food consumption induced by the metabolism of humans.

However, these approaches often lead to inconsistent results (Redlingshofer 2022): balance approaches tend to estimate larger values for food consumption than population-based approaches but rarely distinguish the processes involved. Studies using the balance approach (Bahers et al. 2019; Barles 2009; Hoekman and Blottnitz 2017; Niza et al. 2009; Rosado et al. 2014) apply the DMC concept to estimate the total amount of food used within an area, irrespective of who consumes it (population, food processing companies, etc.) and what the food is used for (including when part of it ends up as food waste). To remedy this shortcoming and provide the necessary information to align the two indicators, Redlingshöfer (2025) has proposed both to disentangle the respective contribution of food consumption and food waste to DMC, and to account for all contributions to food consumption by the population located in an area. Uncertainties in population estimates have indeed made it difficult to draw firm conclusions from the discrepancies in results obtained with the balance approach. In the case of population-based approaches, most studies simply multiply the resident population figures taken from the census, by food availability per capita (Le Noë et al. 2017; Moore et al. 2013; Codoban and Kennedy 2008; Forkes 2007). Only a few studies consider the food consumption of non-resident populations via estimates of the present population to quantify nutrient flows. These concern Brussels (Papangelou et al. 2020), the Paris metropolis or one of its peri-urban areas (Esculier et al. 2019; Tedesco et al. 2017), Amsterdam (Wielemaker et al. 2020), and Indian cities (Boyer et al. 2019). While most of them include residents and commuters or tourists, only Esculier et al. (2019) and Papangelou et al. (2020) include all three. Furthermore, the time spent away from home by residents, which does not contribute to local food consumption, is accounted for only by Esculier et al. (2019) and Tedesco et al. (2017). Finally, none of the studies provide generic formulas to reproduce or adapt these methods to other areas, nor methods to ensure that combining such results at different levels will lead to consistent results.

Besides urban metabolism studies, at least two other research fields require the quantification of food consumption in their analysis: Life Cycle Analysis (LCA) and foodshed studies. LCA, an environmental accounting tool conceptually close to material flow analysis (Finkbeiner et al. 2006), is developing a comprehensive assessment of the environmental impacts of urban sectors and activities at the level of cities, a scale hitherto rarely applied in LCA (Mirabella et al. 2019; Ghaemi and Smith 2020). Few LCA studies at that scale include food consumption (Albertí et al. 2017). Loiseau et al. (2023) developed a method to estimate food consumption at the scale of a city from individual and household consumption data, but they did not integrate the non-resident population. Mirabella and Allacker (2021) presented a bottom-up approach for calculating a population equivalent and characterizing the resident and non-resident population as users of services provided by a city. However, despite the authors' goal of performing inter-city comparisons, this study was limited to the Belgian city of

Leuven, did not provide any generic method applicable to cities in general and required extensive data collection. Foodshed studies, stemming from the geography of food (Kloppenburg et al. 1996; Peters et al. 2009), also require a quantification of food consumption to achieve several goals as outlined by Schreiber et al. (2021): to identify the match between agricultural production and a region's food requirements (expressed globally or by product category, Nixon and Ramaswami 2018); to define the local foodshed size, that is the agricultural land required to cover food consumption needs (Pradhan et al. 2020); and to trace food flows, that is, food movement and concrete organization of the food basin, including food imports and exports (Miller and Mann 2020). Despite the absence of a unified methodology regarding foodshed analysis, it appears that the vast majority of studies base their calculations on the resident population, captured via the statistical system of the countries, states or cities concerned.

1.3. Scientific gap and research question

Overall, few studies integrate non-resident populations in quantitative assessments of an area's food consumption. The contribution of visitors and tourists to food consumption is thus largely overlooked, notably in France and Europe (Panczak et al. 2020). Some authors have suggested that the food metabolism remains under-estimated when only "households" (residents) are considered (Barles 2009; Goldstein et al. 2017) and that using a population-based approach integrating the additional, non-resident population is expected to yield a larger eating population. This would accordingly show a larger annual urban food consumption, compared to simplified approaches based on the resident population. However, the time that residents spend away, on vacation or elsewhere, should also be considered, which makes it unclear whether the net population balance should be positive or negative. Depending on the specific profile of the area, the residents' time spent away (vacation, commuting) may counterbalance the effect of the additional non-resident population on the total eating population and its related annual food consumption. A comprehensive method to assess the present and eating population is therefore necessary to estimate the total food consumption in a delimited area.

This paper aims to develop a conceptual framework to assess the present and eating populations in a given area, drawing on a population-based approach. For applications linked to urban metabolism, the method captures the main phenomena (commutes and tourism) associated with changes in the present population, ensures that the calculated populations are consistent across spatial and temporal scales, and only requires publicly accessible statistical data. By disaggregating the contributions of different types of population or activities, the model provides relevant data to implement sustainability policies adapted to a given area and its specificity.

This paper is organized as follows: first, we present the conceptual framework that relies on publicly accessible statistical data to assess both present and eating populations. We then apply this method across various French administrative scales, revealing (i) significant discrepancies between present, eating and resident populations, and (ii) distinct profiles based on local territorial characteristics. We notably illustrate how the main trends evolve depending on the importance of commute (incoming or outgoing) and tourism in each area. We discuss the usefulness of the approach, its limitations and the implications of these results, notably for research and public policies.

2. Methods

The following section describes the model, data, and the underlying hypotheses used to compute the present and the eating populations. While the method can be applied to any country (provided the required data is available), the examples and analyses in following sections will focus on the case of French administrative areas at different scales. The model provides two alternative metrics: the present population equivalent, which considers population types and their respective time spent in a given area; and the eating population equivalent, which introduces a normalization step to account for the difference in the amounts of food consumed by different populations. A person spending half

their year in an area thus corresponds to a 0.5 population equivalent over said year, while, over a day, a person eating a third of the daily food portion of an adult will contribute to one third of an eating population equivalent over said day, in the area where they consumed the food. As detailed in the discussion, the model is generic and can be adapted to other use cases related to consumption of goods or services associated to the physical presence of individuals.

All code and data used are available on the following git repository: https://codeberg.org/tfardet/article_present-population_Redlingshoefer-Fardet-Petit_2025.

2.1. Present population and present population equivalents

The present population in a given area at a given time can be split into three exclusive categories:

- residential contribution: by residents (r), as given by the national census, during their time spent at their usual residence,
- commuting contribution: by people who commute (c) to their place of work or study in the area (these include residents during their time away from home, if they work within their area of residence),
- touristic contribution: by tourists (t), either nationals on vacation or people from abroad, who spend at least one night in the area.

Over a given time period T, the present population (PP) in the area will be a combination of the populations associated to three categories, indexed by $i \in \{r, c, t\}$, such that P_r gives the residents, P_c, the commuters, and P_t, the tourists. During the week-end, the present population may be mostly composed of the residents and tourists, whereas during business hours on weekdays, the present population may be mostly commuters, senior residents, and tourists. As the time and duration over which these populations are present in the territory vary widely depending on the age-group (indexed j for each age-group – AG), we further divide these three categories by age such that P_{ij} gives the population for category i in the age-group j. To account for these variations in time and space, each sub-population group ij is associated with an amount of time spent ($t_{ij}^{(s)}$) in the area over the full time period T.

From the population and the time spent in the area, we define the present population equivalent in the area, PPE_{ij}, such that

$$\mathrm{PPE}_{ij} = \frac{t_{ij}^{(s)}}{T}\mathrm{P}_{ij} \quad \text{or} \quad \mathrm{PPE}_{ij} = \sum_{sg_i} \frac{t_{sg_i,j}^{(s)}}{T}\mathrm{P}_{sg_i,j} \tag{1}$$

depending on whether group i is a single homogeneous group or is divided into subgroups $\{sg_i\}$.

The population P_{ij}, present for $t_{ij}^{(s)}$ is thus equivalent to a smaller population equivalent PPE_{ij} that would be present during the whole time T. Overall, the total present population (PP) in the area over the time period T is given by

$$\mathrm{PP} = \sum_{i \in \{r,c,t\}} \sum_{j \in \{\mathrm{AG}\}} \mathrm{PPE}_{ij} \tag{2}$$

2.2. Detailed definitions for the population equivalents

In many countries, the national statistics bureau provides data that can help estimate the various populations and the time they spend on the different activities and in any given area. We propose the following definitions for each of the population equivalents, with the parameters and sources for France detailed in Table 1 and in the Supplementary Material Table 3.

The commuter population equivalent is computed as follows:

$$\mathrm{PPE}_{c,j} = \frac{t^{(s)}_{c,j}}{T}\mathrm{P}_{c,j} = \frac{d_{c,j} f^{(t)}_c}{T}\mathrm{P}_{c,j} = \frac{t^{(s)}_{c,j}}{T}\left(\mathrm{P}^{(\mathrm{in})}_{c,j} + \mathrm{P}^{(\mathrm{local})}_{c,j}\right) \tag{3}$$

where $\mathrm{P}_{c,j}$ is the population that leave its residential space to work or study in the area. It can be split into $\mathrm{P}^{(\mathrm{in})}_{c,j}$, the non-resident commuters, and $\mathrm{P}^{(\mathrm{local})}_{c,j}$, the population that both resides and works/studies in the area. The time spent on the trip and daily activities over the total duration T is $t^{(s)}_{c,j}$, which is the number of days where commuting happens, $d_{c,j}$, multiplied by the fraction of the day spent at the commute destination, $f^{(t)}_c$. This is represented by the red bars on Figure 1.

The tourist population is estimated based on data about the nights spent in different accommodations (one "night" is considered as 24h spent locally by one person):

$$\mathrm{PPE}_{t,j} = \frac{1}{T}\left(N^{(m)}_{t,nr,j} + N^{(m)}_{t,n,j} + N^{(nm)}_{t,n,j}\right) \tag{4}$$

with $N^{(m)}_{t,f,j}$ the nights spent by people not living in the country (foreign tourists) in market accommodation, $N^{(m)}_{t,n,j}$ the nights spent by nationals in market accommodation, and $N^{(nm)}_{t,n}$ the nights spent by nationals in non-market accommodation. The latter two, for nationals, are represented by the orange bars on Figure 1. To avoid double counts, each component is computed at the national level, then a fraction is attributed to each subarea based on rules detailed in the Supplementary Material. As both the number of nights spent during friends-and-family stays and their location are among the main sources of uncertainty, we chose to attribute them proportionally to the number of residents in an area. This choice gives a conservative estimate that tends to limit the difference between the resident population and the present population as residents leaving the area are exactly compensated by national tourists visiting other local residents.

Finally, the resident population equivalent is computed as follows:

$$\mathrm{PPE}_{r,j} = \frac{t^{(s)}_{r,j}}{T}\mathrm{P}_{r,j} + \frac{t^{(s)}_{c,j}}{T}\mathrm{P}^{(\mathrm{sedentary})}_{r,j} \tag{5}$$

with the resident population divided into two distinct contributions. The first is common to all residents of age-group j and represents the "standard resident time", $t^{(s)}_{r,j} = t^{(c)}_r + t^{(nc)}_r + t^{(h)}_r$, which is the time spent at home on business days ($t^{(c)}_r$, outside working or studying hours), during no-commute days ($t^{(nc)}_r$) and during holidays ($t^{(h)}_r$, excluding holiday trips that are counted in the tourist population equivalent, as $N^{(m)}_{t,n,j}$ and $N^{(nm)}_{t,n}$). The second contribution is residents who do not commute to a place of work or study on business days. This notably includes retirees and unemployed people. The associated duration $t^{(s)}_{c,j}$ is the same as for commuters. These contributions make up the light and dark blue bars on Figure 1.

In the Supplementary Material we provide alternative formulas to compute population equivalents from employment and education data, for countries where commuter data is not available, or if this data is more precise or granular. We also detail the constraints for time spent by nationals in different activities, to ensure that all their time is accounted for and stays consistent across spatial scales. Based on the graph on Figure 1 A, this consists in making sure that the sum of times spent on all activities is equal to the total time period. In the case of France, official data is provided for all activities except holiday time spent at home ($t^{(h)}_r$) so its value is constrained to T minus the time spent on all other activities.

2.3. Eating population, eating population equivalents, and associated food consumption

For the population that is present in the area over a period of time, one can define another population equivalent associated with the fraction of a daily ration consumed by that population.

This is the eating population (EP), which is defined based on the same principles as the present population (PP). One eating population equivalent is an adult that eats their entire food ration over

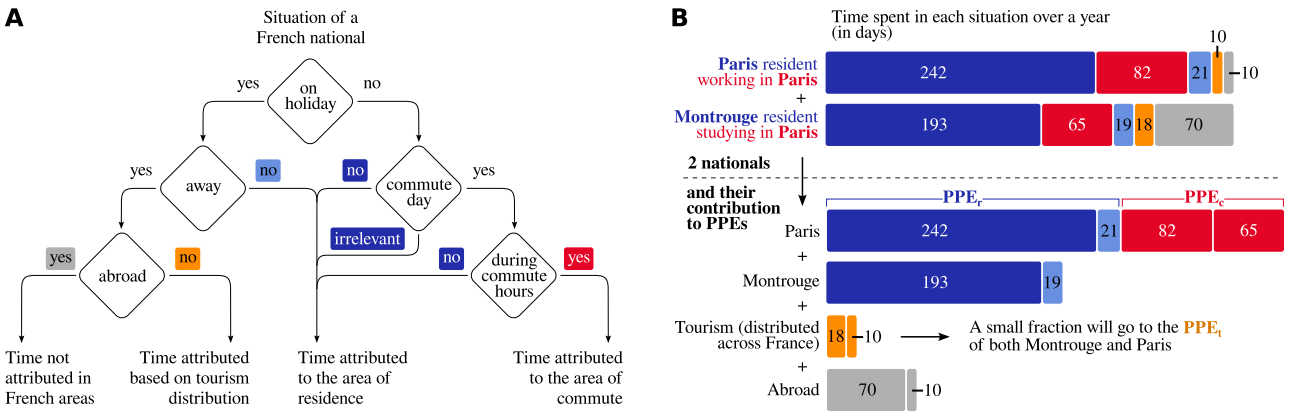


Figure 1: Time spent by nationals in the different situations and their contribution to the present population equivalent (touristic, in orange, residential, in light or dark blue, commuting in red). **A**. A resident can only be in one situation at any given time and each activity is linked to a given area. Thus, ensuring that the sum of the times spent by nationals on each activity is equal to the total time period (as detailed in the Supplementary Material) prevents double counting. Residential time can be during holidays (if the national does not leave their area of residence), during non-commute days outside of holidays (which is all other days for infants and retirees, corresponding to the "irrelevant" case), and outside commute hours on commute days for students and workers. **B**. Example of 2 nationals, one living and working in Paris, another living in Montrouge and studying in Paris, who did a 70-day internship abroad. Each of their situations contribute to the PPE of the area where they live (PPE_r), where they commute (PPE_c) and to the general PPE_t that is then distributed across all French areas according to their touristic potential, as detailed in the Supplementary Material. In this example, both contribute to the PPE_c of Paris, which thus gets a total PPE greater than 1 (the combined time spent there is greater than 1 person-year), whereas the PPE of Montrouge is lower than 1 since the resident does not spend all of their year in Montrouge. Their total contribution to the PPE of French areas is smaller than two person-years since they both spend time abroad.

duration T; each eating population equivalent (EPE) is therefore similar to the associated PPE but uses a food consumed ratio instead of the time spent ratio, and the daily food consumption of the age-group, C_j, comparing it to that of an adult, C_{adult}. In the case of France, this data is based on the INCA 3 study (Anses 2017) which analyzed the food consumption and eating habits of a large panel of inhabitants across multiple regions and food groups.

This leads to the following definitions:

$$\mathrm{EPE}_{t,j} = \frac{C_j}{C_{\mathrm{adult}}} \mathrm{PPE}_{t,j} \tag{6}$$

as tourists consume their entire daily ration in the course of this activity, the eating population equivalent is obtained by multiplying the tourist PPE by the ratio of the age-group food consumption over the adult food consumption.

$$\mathrm{EPE}_{c,j} = \frac{d_{c,j}}{T} \frac{f_c^{(c)} C_j}{C_{\mathrm{adult}}} \mathrm{P}_{c,j} \tag{7}$$

as commuters consume a fraction $f_c^{(c)}$, assumed to be half of their daily ration during lunch and snacking on commute days ($d_{c,j}$) – this value is based on the average distribution of the daily food

Name	Description	Source or value	Level
	Variables		
$\mathrm{P}_{c,j}^{(\mathrm{in})}$	Local workers living outside the area Local students living outside the area	Work commute survey Study commute survey	Local
$\mathrm{P}_{c,j}^{(\mathrm{out})}$	Residents working elsewhere Residents studying elsewhere	Work commute survey Study commute survey	Local
$\mathrm{P}_{c,j}^{(\mathrm{local})}$	Residents working in the area Residents studying in the area	Work commute survey Study commute survey	Local
$\mathrm{P}_{r,j}$	Resident population for age-group j	French national census	Local
$\mathrm{P}_{r,j}^{(\mathrm{sedentary})}$	Residents of age-group j not employed or studying	$\mathrm{P}_{r,j} - \left(\mathrm{P}_{c,j}^{(\mathrm{out})} + \mathrm{P}_{c,j}^{(\mathrm{local})}\right)$	Local
$N_{t,f,j}^{(m)}$	Nights spent in market tourism accommodation by foreign tourists of age-group j	Foreigners' overnight stays	*Département*, or distributed based on local beds
$N_{t,n,j}^{(m)}$	Nights spent in market tourism accommodation by nationals of age-group j	Resident overnight stays	Same as above
$N_{t,n,j}^{(nm)}$	Nights spent in non-market tourism accommodation by nationals of age-group j	See Supplementary Material	National averages, distributed based on houses and residents
	Parameters		
$d_{c,\mathrm{adult}}$	Number of commute days per year for non-student adults (2019)	214 days (INSEE)	National average
$d_{c,\mathrm{students}}$	Number of commute days per year for students (any year until the publication of this work)	175 days (French public service)	National average
$t_{c,j}^{(day)}$	Daily commute and work/study time	9 hours/day (French 2010 schedule)	National average
$t_{c,j}^{(s)}$	Yearly time spent on commute and daily activity	$t_{c,j}^{(day)} \cdot d_{c,j}$	National average

Table 1: Main variables and parameters used to compute the present population and population equivalents in an area. Tourism-related data is at best available at the *département* level; data used at more granular levels (EPCIs, municipalities) is dis-aggregated using specific attribution schemes.

intake across moments of consumption amongst adults in the 18–79 range (Anses 2017).

$$\mathrm{EPE}_{r,j} = \frac{C_j}{C_{\mathrm{adult}}} \left[\underbrace{\frac{d_{nc,j}}{T}\mathrm{P}_{r,j}}_{\substack{\text{full daily consumption} \\ \text{during no-commute days}}} + \frac{d_{c,j}}{T} \left(\underbrace{\frac{(1 - f_c^{(c)}) \left(\mathrm{P}_{r,j} - \mathrm{P}_{r,j}^{(\mathrm{sedentary})}\right)}{8}}_{\substack{\text{fraction consumed} \\ \text{on commute days}}} + \underbrace{\mathrm{P}_{r,j}^{(\mathrm{sedentary})}}_{\substack{\text{full consumption} \\ \text{for sedentary residents}}} \right) \right] \tag{8}$$

as residents consume their whole ration on no-commute days ($d_{nc,j}$), and only $1 - f_c^{(c)}$ on commute days ($d_{c,j}$), except for the "sedentary" residents who work and study and hence eat their whole daily ration in the area of their residence.

As for the present population equivalent, the total eating population equivalent is given by $\mathrm{EP} = \sum_i \sum_j \mathrm{EPE}_{ij}$. The food consumption can then be computed directly from the EP as:

$$\mathrm{FC} = C_{\mathrm{adult}} \cdot \mathrm{EP}. \tag{10}$$

Detailed values for these parameters in France are given in the Supplementary Material.

The main results of this study show the relative difference between the present or the eating population respectively and the resident population, for selected administrative units in France. Relative difference refers to the change between the newly calculated populations and the resident population expressed as a percentage of the original resident population. It is calculated via the following formulas:

$$RD_{\mathrm{PP}} = \frac{\mathrm{PP} - \sum_j \mathrm{P}_{r,j}}{\sum_j \mathrm{P}_{r,j}} \tag{11}$$

for the relative difference between the present population and the resident population, and

$$RD_{\mathrm{EP}} = \frac{\mathrm{EP} - \sum_j \mathrm{P}_{r,j}}{\sum_j \mathrm{P}_{r,j}} \tag{12}$$

for the relative difference between the eating population and the resident population.

2.4. French metropolitan administrative units in 2019 as a case study

Located at the westernmost edge of Europe, metropolitan France covers 551,695 km^2 and had a population of 65.1 million in 2019. Its complex and structured territorial organization encompasses several levels of local authorities: 13 regions (including Corsica), 96 provinces (*départements*), nearly 35,000 municipalities (communes) and 1,236 inter-municipal cooperation areas, abbreviated EPCIs. EPCIs are legal entities formed by multiple municipalities that collaborate to deliver public services. Local authorities have been able to draw up territorial food projects since the 2014 national law for the future of agriculture, food, and forestry established a legal framework for territorial food projects to foster food sovereignty and support local producers and short supply chains. These projects are mainly implemented on an inter-municipal scale, hence the choice of EPCIs as the main level of analysis.

The reference year is 2019, for which a complete data set was available for this study and which was not affected by the impacts of the COVID-19 pandemic on tourism and employment.

3. Results

At the municipal scale, the findings show a negative or close to zero relative difference between present (PP) and resident populations (RD_{PP}, as defined in equation 11) for most of the municipalities, shown by grey to blue areas on the map (Figure 2A). In 70% of French municipalities, the PP is more than 6% lower than the resident population, 6% being the average decrease expected from the resident population leaving on vacation, in France. An RD_{PP} lower than -6% points to the negative contribution of commuters leaving the city to travel to neighboring municipalities for work or study. In 85% of the municipalities, the PP is lower than the resident population ($RD_{\mathrm{PP}} < 0$ on Figure 2B). Municipalities with the highest RD_{PP} (in salmon to red shades) broadly belong to two main categories: touristic areas (mountain regions like the Alps or Pyrenees, the Mediterranean, Atlantic, and Channel coasts, Corsica, and specific recreational destinations such as theme parks) and centers of activity such as large cities that see a large number of incoming commuters.

At the scale of EPCIs (Figure 2C), only around 30% of the areas have an RD_{PP} lower than -6%, and the majority ranges between -5 and 5%, which is a relatively small variation, shown by a

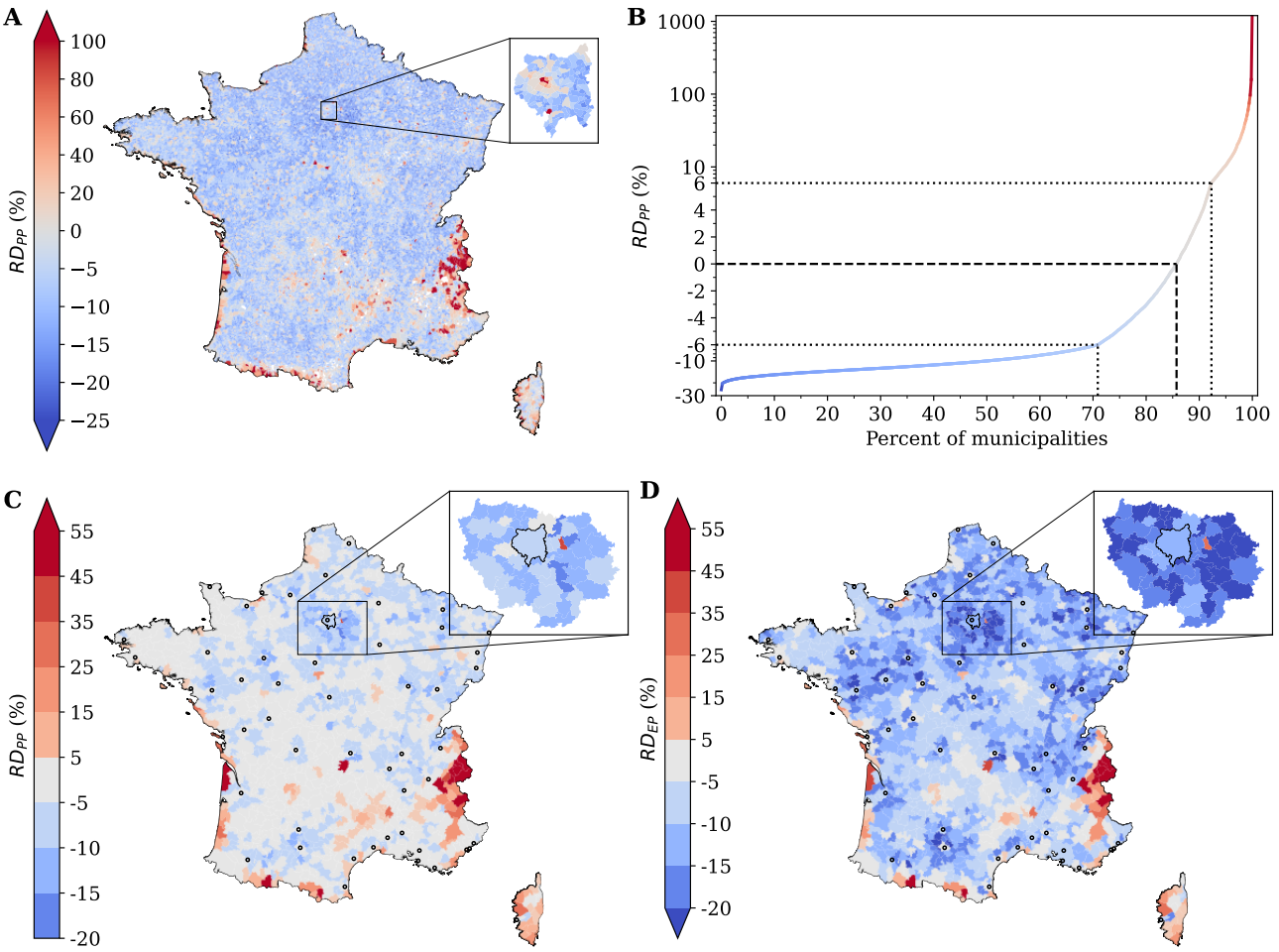


Figure 2: Variation of the population equivalent compared to the resident population, depending on the area. **Top**: Relative difference between the present and resident populations (RD_{PP}, as shown in equation 11) at the municipal level over one year. **A**. Map showing the variation from municipalities that have a lower effective present population compared to the residents, in blue, to municipalities that have a higher present population, in red. **B**. Distribution of the relative difference. The y-scale is logarithmic below -6% and above 6% and linear in-between. A relative difference of -6% refers to the expected decrease from the resident population leaving on vacation. **Bottom**: Relative difference at the level of French inter-municipal cooperation areas (EPCIs) over one year, either between the present (PP) and the resident populations (RD_{PP}, in **C**) or between the eating (EP) and resident populations (RD_{EP}, as given by equation 12, in **D**). A relative difference between -5 and 5% is shown in light grey, while negative changes between -5 and -20% or more go from light to dark blue. Areas where the PP or EP are significantly larger than the resident population are shown in colors going from salmon to dark red. The Greater Paris metropolitan area is marked by a black contour and the inset shows the smaller EPCIs of the Île-de-France region. Black circles designate the centers of the 50 largest urban areas. Note that the color range differs between the municipalities and the EPCIs.

large share of EPCIs in light grey. EPCIs, and especially urban ones, encompass a territory that sustains most of the activities of its population, meaning that fluxes of people in and out of the area represent a smaller fraction of the contribution to the PP compared to individual cities. The 50 largest urban areas in France, shown by a black circle, do not record very high relative differences but instead lie in the majority range, between -5 and 5%. At the temporal scale of a year, incoming population flows (commuters, tourists) are counterbalanced by outgoing flows linked to residents' departures, notably for vacation. EPCIs surrounding large cities usually display a negative RD_{PP},

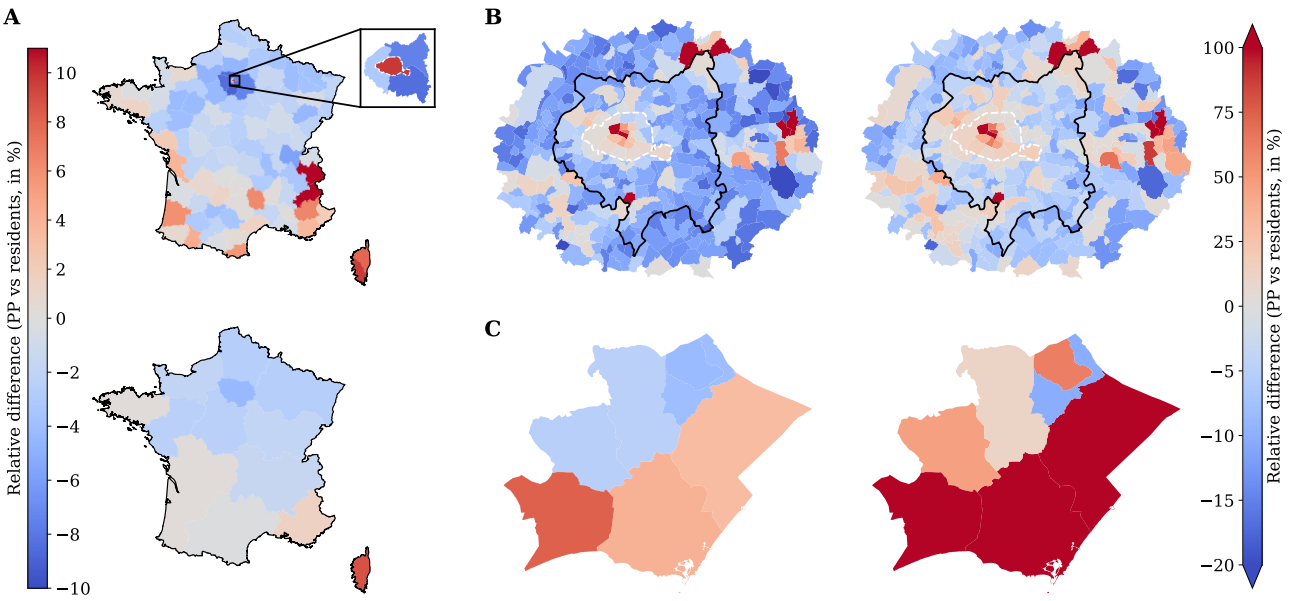


Figure 3: Variation of the gap between the residents and the present populations (PP) depending on the spatial and temporal scale. **A**. Change in the RD_{PP}, as defined in equation 11 at the scale of departments (top) and regions (bottom). **B**. RD_{PP} for cities within the Greater Paris Metropolis (black line) or close to it, over the whole year (left) or a single week (right). The city of Paris is delineated by a white dashed line. **C**. RD_{PP} for the touristic city of Agde and its surroundings over the whole year (left) and during summer time (June–July–August, right).

as the cities drain commuters towards them on working days. In rural areas, far from urban centers, EPCIs display a relative difference close to zero, indicating balanced population flows with other areas, driven by employment and education opportunities within these EPCIs. As for the municipal level, the EPCIs with the highest RD_{PP} are touristic areas that have a small residential population but attract significant numbers of tourists.

While the PP is generally similar to the resident population (most EPCIs are in light grey on Figure 2C), the eating population (EP) is smaller than the resident population in 90% of areas (Figure 2D), with more than 75% of the RD_{EP} in the -5 to -25% range (light to dark blue). This shift can be explained by the fact that children and teenagers consume less food compared to adults, and that commuters have only some of their meals within the area. The only areas where both the PP and EP are substantially larger than the resident population are the touristic areas (from salmon to dark red).

The gap between the residents and the present population (PP) on the territory varies significantly, depending on the spatial and temporal scales. Even at the large spatial scale of French departments, there can still be a discrepancy greater than 10% between the residents and the PP (Figure 3A, top, with a maximum range of -9 to 13%). This decreases at the regional scale but still reaches almost -5% for the Île-de-France region (Paris) and more than 7% for Corsica.

The smaller the spatial scale, the greater these changes will be. For instance, cities within or close to the Greater Paris Metropolis (black line) host populations that differ substantially from their residents alone, both over a whole year (Figure 3B left) and a single week (right). The areas with the largest positive changes (in red, with PP greater than twice the residents) are the areas around CDG airport in the north, Disneyland Paris in the east, the Rungis market in the south, and the center of Paris. The marked difference between yearly (right) and weekly (left) changes shows the importance of choosing the relevant timescale with respect to the question at hand: almost all areas display a larger present population over a week compared to the residents, while most have a present population that is lower than their resident population over the whole year due to the time spent by the residents away from the area during their vacations. Similarly, touristic areas like Agde and the surrounding cities, on the Mediterranean coast, show major changes between the yearly (Figure 3C left) and summer-time

populations (on the right).

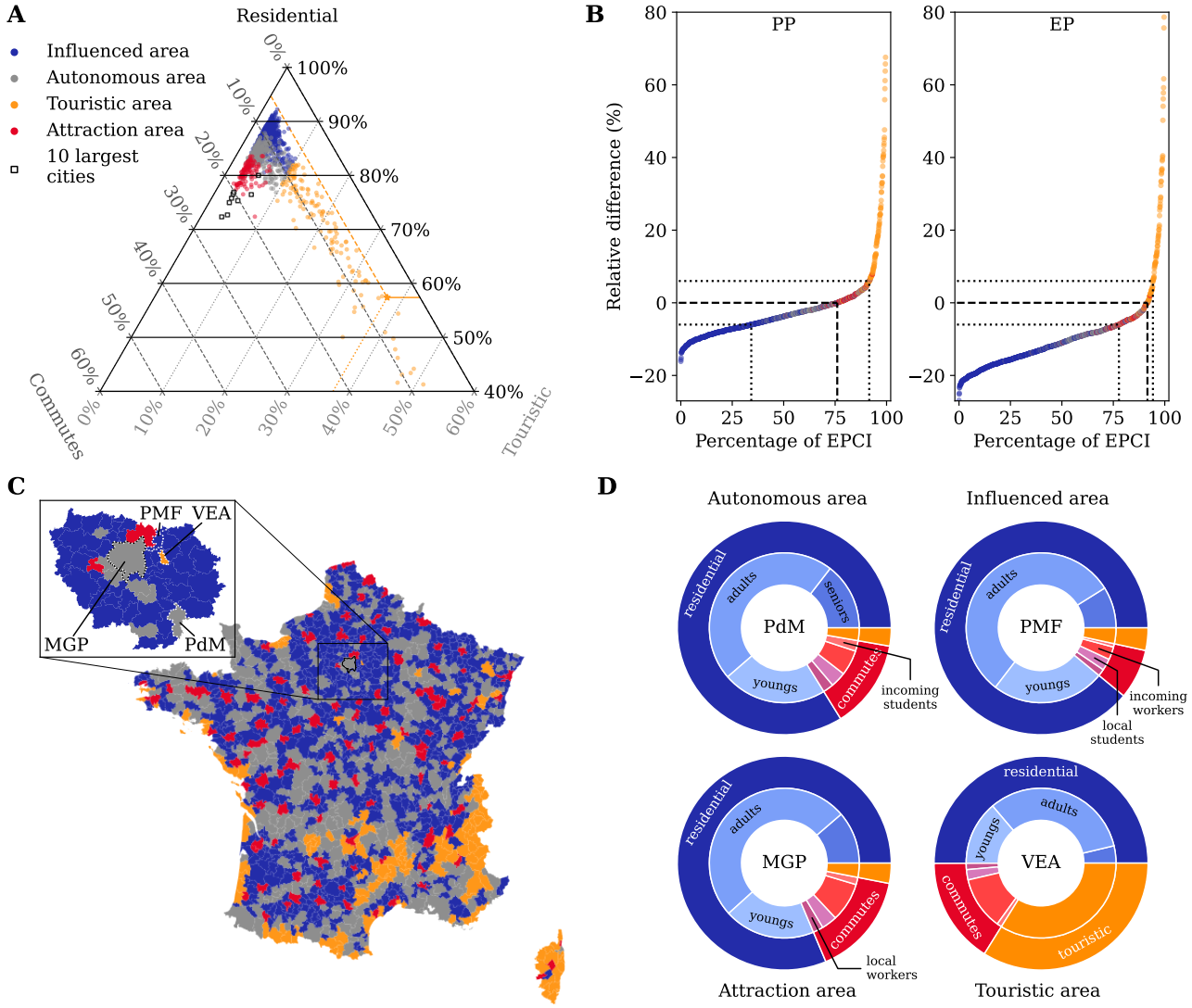


Figure 4: **A**. Ternary plot of the relative contributions from residential, commuting, and touristic activities to the present population (PP) of an area for all French inter-municipal cooperation areas (EPCIs) and the 10 largest cities. The touristic EPCI marked by a star is used as an example to show how to read this diagram: its residents contribute to slightly more than 57% of the PP (full orange line), its commuters contribute to around 5% (dashed orange line), and the tourists contribute to around 37% (dotted orange line). Touristic areas are defined as those receiving more than 10% of their present population from tourism. Attraction areas have more than 10% of the active population coming from outside the territory while influenced areas lose more than 10% of it to other areas. Autonomous areas lie in between. **B**. Distribution of the relative difference between the PP (left) or EP (right) compared to the census population. **C**. Map of the EPCIs classified as attraction, autonomous, influenced, and touristic areas. **D**. Detail of the specific contributions of different populations to the PP for each of the four types of territory (highlighted by dotted white contours on the inset in **B**). MGP: Métropole du Grand Paris (Greater Paris Metropolis); PdM: Pays de Montereau; PMF: Plaine et Monts de France; VEA: Val d'Europe Agglomeration (where Disneyland Paris is located).

The contribution of the various population equivalents to the present population (PP) led us to classify the areas into four main categories (Figure 4A), depending on the relative contribution of each population to the PP — see Supplementary Material for more details about the precise classification criteria. Touristic areas are characterized by a touristic contribution to the PP that is larger than

10%. Autonomous areas have a balanced mix of population equivalents, whereas the two remaining areas are characterized by the commuter population equivalents: attraction areas are characterized by an incoming commuter population that accounts for more than 10% of the active population; and areas under influence see more than 10% of their active population leave to work or study elsewhere.

Most "influenced areas" have a PP that is at least 6% lower than its resident population, meaning that the decrease in the number of individuals that are effectively present in the area is higher than what can be explained by residents going on vacation. Almost all of them (98%) have a negative RD_{PP}, that is, a present population that is smaller than the resident population (Figure 4B and supplementary Figure A1). The majority of autonomous areas have a balanced RD_{PP}, between -6 and 0%, some with a stronger touristic and commuter contribution that places them in the positive range. All attraction areas offset the departures of their residents on vacation ($RD_{\mathrm{PP}} > -6\%$) through large influxes of commuters, to the extent that most (75%) have a positive RD_{PP}. Finally, the touristic areas are the only ones that consistently show population gains over a year, with $RD_{\mathrm{PP}} \geq 0$ for 97% of them. They are also the only type of territory where most areas (70%) have an eating population that is larger than the resident population.

4. Discussion

4.1. Significance of the results

We provide a consistent framework, based on publicly available and periodical demographic data, that can be used to assess the present population over a certain period of time in an area. The concepts and data used are a common standard in an international context, allowing for replication in any country where such data is available. This method, based on population equivalents rather than individuals, enables one to account for the variety of populations that spend time in the area (residents, commuters, tourists) and the different ways and periods in which this area is used (residential, occupational/educational, or touristic use). Compared to previous studies, we provide detailed formulas for the population equivalents and discuss the different sources of data that can be used to compute them (Supplementary Material D). We also discuss how these methods ensure that the calculations are consistent across all geographical scales, as well as the self-consistent equation that must be verified by the touristic contribution of nationals to avoid double-counts or missed contributions (Supplementary Material A). Because of the model's generic formalism, other types of population equivalent can easily be proposed to answer the need of other fields. For instance, other socio-metabolic studies dealing with sanitation (the downstream part of food consumption, Starck et al. 2024) may adapt this framework to analyze nutrient flows from human excretion at smaller scales, using fractions based on daily excretion patterns instead of time or food consumption fractions. These results could then be compared to the standard "population equivalent" used to dimension wastewater treatment plants.

Using this population-based approach, we show that, at the scale of municipal and inter-municipal areas in France, most locations have a present or eating population that is lower than their resident population (at least 75% of areas for the PP and up to 90% for the EP). Since annual food consumption is directly proportional to the EP, this also means that most areas have a lower food consumption compared to estimations in previous studies that used the resident population. This directly contradicts the oft-formulated hypothesis (Goldstein et al. 2017; Barles 2009; Niza et al. 2009) that such calculations underestimate the consumption because they do not account for population flows, and in particular for additional non-resident population such as commuters or tourists, in large urban areas. Since the time that residents spend away from the area — e.g. for vacation or family visits, so far mostly ignored in previous studies — is the main contributor to the reduced present population, this also means that studies that overlook this factor risk over-estimating food consumption in the area, especially if they consider additional incoming populations like commuters or tourists. It is important to note that the results shown in the main text use the most conservative distribution for the contribution of nationals on holiday, that is, the one that minimizes the discrepancy between the present and the resident populations. This means that the actual error made using the resident population could

actually be 5 to 10% larger in some areas, as shown in Figure A2 of the Supplementary Material.

Territory	**Year**	**P_r**	**EP**	**FC**	**DMC**	**DMC/EP**	**Source**
Rennes Métropole	2012	408,549	384,162*	274	1,277	3.53	Bahers and Giacchè (2018)
Le Mans Métropole	2012	184,446	177,902*	127	625	3.28	Bahers et al. (2019)
Île-de-France	2003	11,259,000	10,100,000†	7,200	10,300	1.04	Barles (2009)
Paris	2003	2,166,000	2,291,000†	1,632	3,870	1.69	Barles (2009)
PC	2003	4,155,000	3,527,000‡	2,511‡	2,130	0.6	Barles (2009)
PPC	2003	6,321,000	5,818,000†	4,143	6,000	1.03	Barles (2009)
PPC	2014	6,754,282	6,202,964*	4,424	4,997	0.8	Redlingshöfer (2025)

Table 2: Comparison between resident (P_r) and eating (EP) populations, as well as food consumption (FC) — calculated by means of the new model — and previous DMC calculation for food and agricultural products, using material flow analyses for five territories in France. Food consumption is computed by considering all INCA 3 groups except tapwater and hot beverages ($C_{\text{adult}} = 1,952$ g/day). PPC is "Paris Petite Couronne", the combination of Paris and "Petite Couronne" (PC), the three surrounding *départements*, numbered 92, 93 and 94. FC and DMC are expressed in kilotons, DMC/EP is in ton per capita per year. *Tourism data estimated from 2015 and 2018 values. †EP estimated from P_r using the 2019 value of RD_{EP}. ‡Values for PC are computed as the difference between PPC and Paris for consistency.

This comprehensive population-based approach also provides further insights to complement consumption indicators in material accounting methods (European Commission. Statistical Office of the European Union. 2018). A comparison of the work of Barles (2009), Bahers and Giacchè (2018), Bahers et al. (2019), and Redlingshöfer (2025) with the results of this study (Table 2), yields insights into the contribution of food consumption to DMC. Using the eating population (EP) to normalize the DMC also provides further insight into the metabolic profile of the areas. The first two studies used the broader category of agricultural biomass, including animal feed or industrial material in addition to food, while Redlingshöfer (2025) restricted the data considered to food flows only. In the case of Paris and the surrounding areas, however, biomass flows are mostly associated to food consumption, as mentioned in Barles (2009).

In the case of the two intermediate French metropolises, Rennes and Le Mans, (Bahers et al. 2019), there is a large gap between food consumption and DMC for biomass. Both areas include livestock and wood-fired power plants, which may explain most of the remaining biomass consumption. Other transformations of wood and agricultural biomass, not only produced in large farmland areas on the outskirts of both cities (respectively 34% and 55% of the EPCI of Le Mans and Rennes consist of farmland) but also imported, would be another area for investigation.

In the case of the Paris metropolitan area and Île-de-France region, however, both livestock and biomass plants were mostly absent in 2003, meaning that human food consumption represented most of the biomass flows for (Barles 2009). The predominance of food in the local biomass flows is confirmed both by the smaller difference between food consumption and DMC for Paris Petite Couronne (PPC) and Île-de-France and by the limited discrepancy, for PPC, between the values including all biomass in 2003 and those using only food in 2014. Both PPC and Île-de-France show very similar metabolic

profiles once the DMC is normalized by the eating population (1.03 and 1.04 t/cap/y respectively), whereas Paris and PC show vastly different profiles with respectively close to 50% increase and decrease in apparent food consumption per capita (1.69 and 0.6 t/cap/y respectively).

Since our model accounts for most contributions of temporary populations, we can deduce that, if these figures are correct, this apparent consumption can only be explained by excursionists (same-day visitors) from the suburbs who come to Paris to eat, e.g. for dinner on weekdays or for any of their daily meals during week-ends. Considering that each eating population equivalent in Petite Couronne (PC) should also have a DMC/EP of 1.03 t/cap/y (the same as PPC and Île-de-France), excursionists from PC moving to Paris to eat should thus be responsible for the missing 0.43 t/cap/y. This consumption from excursionists represents 1,516 kt/y of DMC, which would indeed resolve the discrepancy of Paris' metabolic profile compared to other areas by bringing the consumption of its eating population down to 2,354 kt/y, which gives a DMC/EP of 1.03 t/cap/y, the same as that of the other areas.

The use of the eating population as a normalization factor thus enables us to better highlight and discuss variability in the metabolic profiles of different areas. Given the population accounting in the model, we can state that the inconsistent values for Paris and PC cannot be explained by tourists or commuters. This leaves two possible explanations: either there are issues with the data, with flows being incorrectly attributed to Paris instead of surrounding areas, or there is significant excursionist behavior, suggesting that many commuters from PC may stay in Paris for dinner and that many PC residents eat in Paris on week-ends.

Finally, the small difference between the estimated food consumption and the food DMC computed by Redlingshöfer (2025) for PPC in 2014, validates the results of our model: the difference between the estimated food consumption and the food-related DMC leads to approximately 573 kt of food waste, close to the 679–783 kt range from the article. The remaining gap may come from a variety of sources, including the fact that the consumption values come from a national average and the use of estimated data for tourism.

4.2. Policy relevance

The population-based approach developed in this study is useful for policy-making, particularly in the field of food systems, as policymakers are increasingly re-engaging with food-related issues. To implement territorial food projects, local authorities lack methods and tools to conduct a quantitative assessment of their food system as an initial step. In this regard, open-access, open-source tools, such as the one developed by the authors for the French case, enable local authorities to quantify food consumption at the territorial scale. Strategies for strengthening cities' self-sufficiency and reconnecting production and consumption can benefit from this study's approach (Tedesco et al. 2017). Doernberg et al. (2019) highlighted that sectoral approaches to urban food strategies prevail over integrated ones and proposed a framework to analyze policy action and instruments, with a focus on urban food policies. To complement this framework, our quantitative approach aims at characterizing the different categories of population responsible for food consumption within a given area. The composition of the present population, and the eating population in the case of food, provides useful insights to design targeted policies for the socio-ecological transition in the field of food, and potentially also for the public service sector (energy, transport, etc). By analyzing population types and use patterns of a given area, our framework categorizes territories into four distinct profiles: influenced, touristic, attraction, and autonomous areas. Attraction and touristic areas, characterized by a comparably high number of commuters and tourists, can use restaurants and canteens to offer locally-sourced food services or set up waste reduction initiatives. Influenced areas, on the other hand, may want to focus on the specific needs of busy commuters who often resort to practices that are mostly wasteful and unsustainable (Dobernig and Schanes 2019; Schanes et al. 2018).

4.3. Limitations

The main limitation of this study stems from the need to dis-aggregate data at local scales (especially for EPCIs and municipalities) since some of the parameters used are national averages or estimates that may not precisely account for the specificity of local areas. For the time spent at the usual place of residence and at the place of work or study, we used national averages that may display variations depending on the local socio-economic context. For the contribution of tourism, we made assumptions about the distribution of overnight stays from data available at national and *département* scale that may not precisely reflect the actual choice of accommodation by local tourists. Furthermore, tourism data may be incomplete for overnight stays at friends and family, as well as for nights spent in platform-based tourism, such as Airbnb. Finally, statistical data cannot account for excursionists, which may also have a significant contribution at small spatial scales, especially close to large metropolises, as seen in the case of Paris and the Petite Couronne. This issue can be alleviated by considering territories that encompass the typical living area of local residents, e.g. considering Paris and the Petite Couronne together in the previous case.

The main source of uncertainty that was identified, based on sensitivity analysis, was related to overnight stays at friends and family, which is the largest share of tourism in France (cf. Supplementary Material). This may not be accurately captured by surveys of the French statistics agency; it is estimated at national scale only, and there was no available information about where these overnights stays take place. Three different distribution rules were tested, according to: a) the resident population (this is the rule used to compute the results in the main text); b) the number of secondary houses in the area; and c) the local capacity of commercial accommodation. Sensitivity analysis showed that, in 90% of EPCIs, changing the distribution rule led to an error in the evaluation of the present population (PP) that was lower or equal to 10%. However, in the remaining EPCIs, the estimated PP could vary by up to 40% (see supplementary Figure A3). Furthermore, the choice of rule had a significant impact on the sensitivity of the PP to the precise value of the number of nights spent at friends and family: the resident population rule was almost completely insensitive to the exact number of overnight stays, whereas the other rules led to notable variations in the local PP estimation. A similar issue was encountered for nights spent in commercial accommodation at scales smaller than French departments, where detailed data is not available. To overcome this issue, we chose to distribute these overnight stays according to the local capacity of commercial accommodation, though this may not match the exact local occupation rate.

4.4. Contribution to population studies

Population studies and related fields have underscored the challenge of estimating how many people are present and for how long in a given area over a given time (Panczak et al. 2020). Building on Smith's concept of temporary populations (Smith 1989), our work contributes to meeting this challenge while incorporating socio-metabolic considerations (Haberl et al. 2019). We provide a comprehensive framework and novel metrics of population equivalents, which are the present population and the eating population equivalents, adjusted to the context of food systems in small-scale areas. Based on two main pillars — distinguishing between population types, based on age and occupation (workers, students, retirees...), and considering their temporal presence in a given area — our work resonates with current trends in population studies.

Driven by big data about people's use of time and resources, a current trend lies in exploring how to translate this data into temporary population estimates. ICT-based data, for instance, offers promising perspectives for refining the understanding of mobility patterns, as recently demonstrated by Lévy et al. (2023) with mobile-phone tracking in the case of France. Our findings on present populations are qualitatively similar to the geography of inhabitants described by these authors, particularly for touristic areas and urban centers.

Our study seeks to prioritize data accessibility by using public administrative surveys to model populations. This choice allows us to remedy the representation biases inherent in mobile data, such

as data anonymization or partial coverage of population. People who may not be technologically equipped, such as children, the elderly and vulnerable groups, are often underrepresented or misidentified (Vallée and Lenormand 2024). Furthermore, mobile phone data does not contain any relevant socio-demographic information about individuals or the reasons for their movements, which makes it difficult to use when estimating food consumption. The use of open data therefore facilitates access for policy makers and allows all demographic categories to be included, regardless of the devices they own.

Although our work is driven by the challenge of modeling an area's food metabolism and developing appropriate metrics, population studies and related fields can benefit from our framework and further refine it through additional data sources.

5. Conclusions

Providing a standardized and actionable method to estimate population equivalents is an important step for socio-metabolic research. The comprehensive population-based approach presented in this study provides a more accurate approach than previous studies to characterize urban food consumption and analyze which populations are involved. It also shows that territorial analyses that rely on population data to assess specific material flows should take the discrepancy between the present population and the resident population into account, especially for touristic and metropolitan areas that can display major seasonal variations and large population movements.

Applicable to any administrative unit, from small municipalities to large regions, this framework enables bottom-up estimations of some activities, such as part of a city's energy demand, environmental footprints, or nutrient flows, that depend on the time spent by people in the area. Given the publicly available census data used, the framework can be adjusted to countries with similar legal data structures available from national statistics authorities. Standardized guidelines for the present and eating population concepts, in collaboration with the socio-metabolic research community, would facilitate comparison between studies. Per capita material or energy flows could be normalized by the present population or one of its variants to provide a deeper understanding of a city's material use by limiting the impact of population changes as a source of variation.

The presented framework opens up a range of possibilities for future research, including physically consistent modeling of diet shift, reduced mobility, and nutrient recovery scenarios, as well as for designing urban demand-side policies towards more sustainable cities.

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A. Consistency of the time spent by nationals in the different activities

A.1. Time consistency

The total time T is split into two main periods:

- the time spent on touristic activity, $t_t^{(s)}$,

- the days where the resident is present in the area (at least in the morning and the evening), d_p, split into "commuting" and "non-commuting" days, $d_p = d_c + d_{nc}$

with $T = t_t^{(s)} + d_p$, and $d_p = t_r^{(s)} + t_c^{(s)}$, such that:

$$T = t_t^{(s)} + \underbrace{\left(d_{nc} + \frac{d_c}{2}\right)}_{t_r^{(s)}} + \underbrace{\frac{d_c}{2}}_{t_c^{(s)}}. \tag{13}$$

The days dedicated to touristic activity are further split into main parts, based on data about nights spent in different locations (one "night" is considered as 24h spent locally by one person):

- nights spent abroad, $N_{t,r}^{(out)}$,
- nights spent in market accommodation in the country, $N_{t,r}^{(m)}$,
- nights spent in non-market accommodation in the country, $N_{t,r}^{(nm)}$,

with non-market accommodation being further split into nights spent in secondary housing, $N_{t,r}^{(sh)}$, or at friends and family, $N_{t,r}^{(f)}$. The average time spent on holiday for one person is thus given by:

$$t_{t,r}^{(s)} = \frac{N_{t,r}^{(out)} + N_{t,r}^{(m)} + N_{t,r}^{(sh)} + N_{t,r}^{(f)}}{\mathrm{P}_{\mathrm{legal}}} \tag{14}$$

If we define the total number of person-days that are not spent on commutes as $D_{nc} = \sum_j (T - d_{c,j})\mathrm{P}_{\mathrm{legal},j}$, then for a given population group j, the amount of nights spent in a specific tourist accommodation can be approximated by:

- $N_{t,r,j}^{(out)} = N_{t,r}^{(out)} \frac{\mathrm{P}_{\mathrm{legal},j}}{\mathrm{P}_{\mathrm{legal}}}$,
- $N_{t,r,j}^{(m)} = N_{t,r}^{(m)} \frac{\mathrm{P}_{\mathrm{legal},j}}{\mathrm{P}_{\mathrm{legal}}}$,
- $N_{t,r,j}^{(sh)} = N_{t,r}^{(sh)} \frac{\mathrm{P}_{\mathrm{legal},j}}{\mathrm{P}_{\mathrm{legal}}}$,
- $N_{t,r,j}^{(f)} = N_{t,r}^{(f)} \frac{(T - d_{c,j})\mathrm{P}_{\mathrm{legal},j}}{D_{nc}}$.

Since minors cannot be involved in touristic activities except those at friends or family without a supervising adult, all nights except those spent at friends or families are distributed according to the population share, while the latter are distributed according to the total person-days for each age group.

We then define the time spent on touristic activities for nationals:

$$t_{t,j}^{(s)} = \min\left(T - d_{c,j}, \frac{N_{t,r,j}^{(out)} + N_{t,r,j}^{(m)} + N_{t,r,j}^{(sh)} + N_{t,r,j}^{(f)}}{\mathrm{P}_{\mathrm{legal},j}}\right) \tag{15}$$

Overall, this guarantees that the time spent by any group always adds up to T for nationals:

$$T = t_{t,j}^{(s)} + t_{r,j}^{(s)} + t_{c,j}^{(s)} \tag{16}$$

Since all durations except d_{nc} are constrained by available data, consistency is ensure by defining:

$$d_{nc,j} = T - t_{t,r,j}^{(s)} - d_{c,j}. \tag{17}$$

A.2. Consistency of the present population

Since it is not possible to obtain the age groups associated to the nights spent in an area, nationals' nights are distributed proportionally across all groups while all foreign tourists nights are associated to the adult population, such that:

$$\mathrm{PPE}_{t,j} = \mathrm{PPE}_{t,r,j} + \mathrm{PPE}_{t,nr,j} = \underbrace{\frac{N^{(m)}_{t,r,j} + N^{(nm)}_{t,r,j}}{T}}_{\mathrm{PPE}_{t,r,j}} + \begin{cases} \dfrac{N^{(m)}_{t,f}}{T} & \text{if } j = \text{adult} \\ 0 \text{ otherwise} \end{cases} \tag{18}$$

Combined with the time consistency, the present populations and the contributions from people working or going on holiday abroad always sum up to the total legal population:

$$\underbrace{\mathrm{PPE}_{r,j}}_{\substack{\text{residents} \\ \text{at home}}} + \underbrace{\mathrm{PPE}_{c,j}}_{\substack{\text{residents commuting} \\ \text{in the country}}} + \underbrace{\mathrm{PPE}_{t,r,j}}_{\substack{\text{residents' holiday} \\ \text{in the country}}} + \underbrace{\frac{t^{(s)}_{c,j}}{T}\mathrm{P}^{(\mathrm{out})}_{c,j}}_{\substack{\text{residents} \\ \text{commuting abroad}}} + \underbrace{\frac{\mathrm{N}^{(out)}_{t,r,j}}{T}}_{\substack{\text{residents on} \\ \text{holiday abroad}}} = \mathrm{P}_{\mathrm{legal,j}} \tag{19}$$

B. Specific data for France

B.1. Age groups and schedules

The age groups and schedules considered for France are:

- babies (less than 3 years-old), who are considered spending all their non-touristic time in the area of residence,
- kids (3–10 years-old), spending 175 days a year at school,
- teenagers (11–17), also spending 175 days a year at school,
- adults (18–64), spending 211 days at work, or 175 days in higher education for students,
- seniors (above 65), who are considered spending non-touristic time at their place of residence.

B.2. Calculation of tourists

In France, the most precise level for tourist nights is the department (2nd administrative level). To extrapolate tourist nights at a more granular scale, e.g. for municipalities, we use the following formula:

$$N^{(m)}_{t,\mathrm{local}} = \frac{C^{(m)}_{t,\mathrm{local}}}{C^{(m)}_{t,\mathrm{dep}}} N^{(m)}_{t,\mathrm{dep}} \tag{20}$$

where the fraction of the department-level nights $N^{(m)}_{t,\mathrm{dep}}$ retained as local nights $N^{(m)}_{t,\mathrm{local}}$ is given by the ratio of the local capacity $C^{(m)}_{t,\mathrm{local}}$ (e.g. number of rooms for hotels) and the total capacity for the department $C^{(m)}_{t,\mathrm{dep}}$.

The number of nights spent by nationals in non-market accommodations locally is computed via:

$$N^{(nm)}_{t,r,\mathrm{local}} = \frac{R_{s,\mathrm{local}}}{R_{s,\mathrm{France}}} N^{(sh)}_{t,r} + \frac{\mathrm{P}_{\mathrm{legal,local}}}{\mathrm{P}_{\mathrm{legal,France}}} N^{(f)}_{t,r} \tag{21}$$

R_s the number of secondary residences, is used as a proxy for the distribution of nights spent in secondary housing ($N^{sh}_{t,r}$), and the legal population is taken as a proxy for the distribution of nights spent at friends and family ($N^{(f)}_{t,r}$).

The total number of local tourist nights is thus given by:

$$N_{t,\text{local}}^{(\text{total})} = N_{t,\text{local}}^{(m)} + N_{t,\text{local}}^{(nm)} = \frac{C_{t,\text{local}}^{(m)}}{C_{t,\text{dep}}^{(m)}} N_{t,\text{dep}}^{(m)} + \frac{R_{s,\text{local}}}{R_{s,\text{France}}} N_{t,r}^{(sh)} + \frac{\text{P}_{\text{legal,local}}}{\text{P}_{\text{legal,France}}} N_{t,r}^{(f)} \tag{22}$$

with the description of the parameters given in Table 3.

Name	Description		Value	Source
$N_{t,r}^{(out)}$	Nights spent by nationals abroad:	2019	$272 \cdot 10^6$	EU + NON EU
		2014	$235 \cdot 10^6$	Corrected Eurostat 2015 data
$N_{t,f}^{(m)}$	Nights spent in market tourism accommodation by foreign tourists		local	foreigners' overnight stays (market)
$N_{t,r}^{(m)}$	Nights spent by nationals in market accommodation:	2019	$305 \cdot 10^6$	Hotel + Camping + Others
		2014	$276 \cdot 10^6$	Camping 2014 + Others 2014
$N_t^{(m)}$	Nights spent in market tourism accommodation by all tourists		$N_{t,r}^{(m)} + N_{t,f}^{(m)}$	
$N_{t,r}^{(sh)}$	Nights by nationals in secondary housing:			
		2019	$126 \cdot 10^6$	INSEE data
		2014 (= 2018)	$131.9 \cdot 10^6$	
$N_{t,r}^{(f)}$	Nights by nationals at friends or family:			INSEE data for
		2019	$410 \cdot 10^6$	"friends/family" + "other
		2014 (= 2018)	$460 \cdot 10^6$	non-market"

Table 3: Parameters used to compute the number of tourist overnight stays for metropolitan France. Note that the 2014 values needed for the discussion were not available so estimates were used instead. For nights spent abroad, the Eurostat data from 2015 was used; however, since the 2019 Eurostat data did not match the French value in 2109, we multiplied the Eurostat 2015 data by $N_{t,r}^{(out,2019)}/Eurostat(2019)$; for nights in secondary housing or at friends and family, data from 2018 was used.

C. Classification of the territories

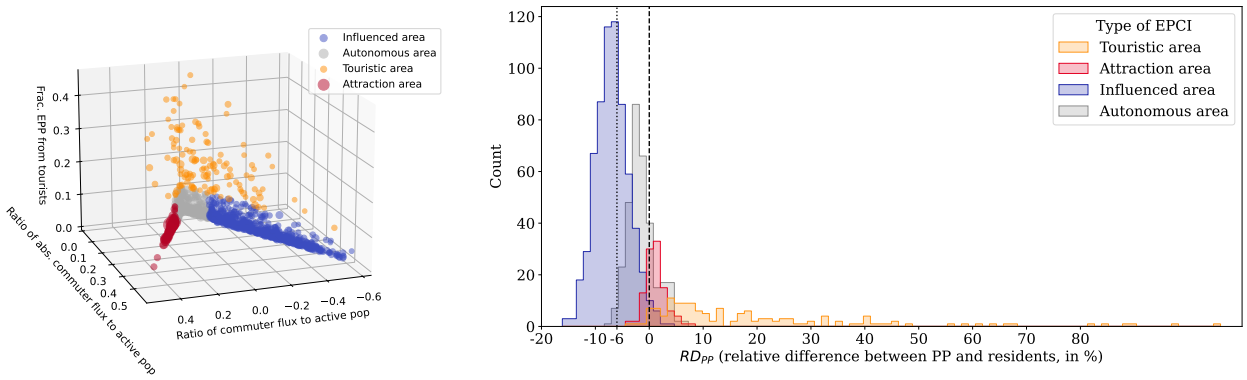


Figure A1: Classification of the territories into 4 groups. **Left**: 3D plot showing the contribution of tourists to the PP and the absolute or signed ratio of commuters to the active population. Both values are used to discriminate the 4 types of territories. **Right**: Distribution of RD_{PP} by type of territory.

D. Alternative definitions or data for the population equivalents

D.1. Using different data sources

Depending on the data available, the different population types can be constructed independently of commuter data, using (un)employment, retirement, and education data instead.

The commuter population, which is effectively people working and studying locally (we will call them "active", here), can equivalently be computed as:

$$\mathrm{PPE}_{c,j} = \mathrm{PPE}_{a,j} = \frac{t^{(s)}_{c,j}}{T}\mathrm{P}_{a,j} \tag{23}$$

with $\mathrm{P}_{a,j}$ being the "active" population, defined here as the sum of the working and studying populations for age group j.

The legal population stays

$$\mathrm{PPE}_{r,j} = \frac{t^{(s)}_{r,j}}{T}\mathrm{P}_{\mathrm{legal},j} + \frac{t^{(s)}_{c,j}}{T}\mathrm{P}^{(\mathrm{sedentary})}_{r,j} \tag{24}$$

with

$$\mathrm{P}^{(\mathrm{sedentary})}_{r,j} = \mathrm{P}_{\mathrm{legal},j} - \mathrm{P}_{a,j}. \tag{25}$$

D.2. Using non-positive contributions in intermediate "population equivalents"

To make definitions simpler, at the detriment of having them positive, one can also define the total population equivalents as a sum of other intermediate contributions that are not independent.

The residential contribution is computed using all present days, including those where some commute happens:

$$\mathrm{PPE}^{*}_{r,j} = \frac{d_p}{T}\mathrm{P}_{\mathrm{legal},j} \tag{26}$$

To compensate, the commuting contribution is given by:

$$\mathrm{PPE}^{*}_{c,j} = \frac{t^{(s)}_{r,j}}{T}\left(\mathrm{P}^{(\mathrm{in})}_{c,j} - \mathrm{P}^{(\mathrm{out})}_{c,j}\right) \tag{27}$$

where resident leaving the area ($\mathrm{P}_{c,j}^{(\mathrm{out})}$) during commute time have a negative contribution, which can lead to $\mathrm{PPE}_{c,j}^{*} < 0$.

The touristic contribution remains unchanged:

$$\mathrm{PPE}_{t,j}^{*} = \mathrm{PPE}_{t,j} \tag{28}$$

This leads to:

$$PP = \sum_{j} \mathrm{PPE}_{r,j}^{*} + \mathrm{PPE}_{c,j}^{*} + \mathrm{PPE}_{t,j}. \tag{29}$$

E. Sensitivity analysis

E.1. Distribution of visits to friends and family

The main source of tourism-related population movement available in statistical data is nationals visiting friends or family. Unfortunately, the amount and destination of these visits is unknown and changes in the way these fluxes are distributed is the main source of uncertainty in the calculation of the present population (PP) for some areas.

We consider three ways in which nights spent at friends and family could be distributed:

- as friends and family are residents, these overnight stays might simply be dispatched according to the population in an area (i.e. the number of residents) – this is the assumption that was used in the main text;
- since hosting people requires space, these overnight stays might preferably be spent in places where these friends and family have secondary houses;
- finally, people may preferably visit friends and family that are in areas of touristic interest.

Of course, one could imagine an infinite number of ways to distribute these overnight stays, but these three possibilities are already different enough that they provide an idea of how significantly such changes could affect the PP. Furthermore, the real distribution is likely to be explain at least partly by each of these three contributions.

Though the distributions remain qualitatively similar at both communal (Figure A2) and EPCI levels (Figure A3), there can be significant quantitative variation in the PP of an area depending on the chosen hypothesis. Indeed, even at the EPCI level, though most of them (roughly 90%) display lower than 10% variations, especially for areas with more than 10^4 PP, some results may change by up to 40% for touristic areas depending on the hypothesis for nights spent at friends and family.

E.2. Remote work

Compared to the hypothesis on tourism, remote work plays a much smaller role except in very specific areas. As shown in Figure A4, even under a highly unrealistic hypothesis of 60% of the whole population working remotely 3 days a week, 95% of the communes do not display relative errors higher that 5% in the PP calculation.

F. Geography and population of Metropolitan France

Located at the westernmost edge of Europe, France, in its metropolitan part, covers 551,695 km2. The territory features varied yet generally moderate topography. Metropolitan France has a population of 66.1 million (as of January 2024). The distribution of the French population across mainland territory shows significant disparities: concentration around major cities, increasing attractiveness of coastal

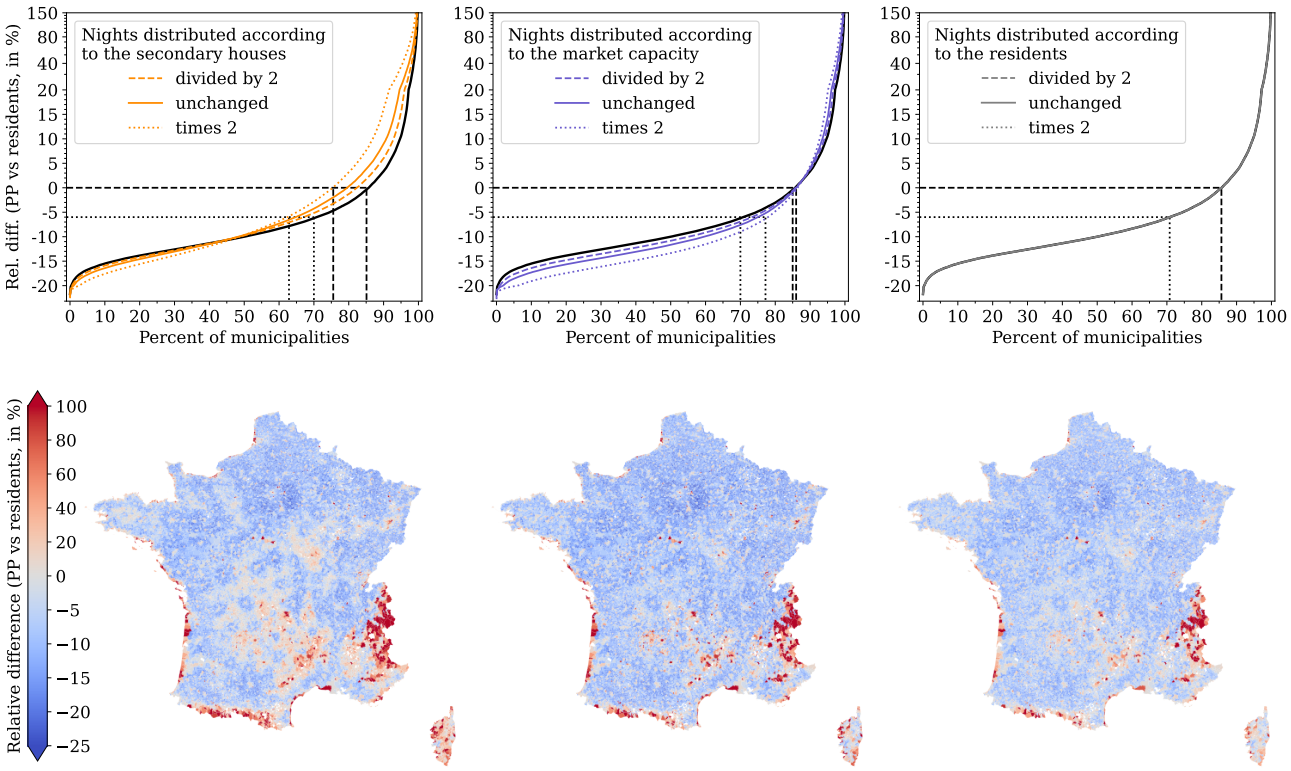


Figure A2: Sensitivity of the PP to the number of nights spent at friends and family and the way they are distributed spatially at the communal level. Left: impact of changing the distribution of overnight stays to follow that of secondary houses on the distribution of relative differences (top) and on the associated map (bottom). Center: distribution of overnight stays according to market capacity. Right: distribution of overnight stays according to the residents. The number of overnight stays is also varied: either divided by two (dashed lines) or multiplied by two (dotted lines). The maps represent the situation for the initial number of overnight stays.

and southern regions, and low-density areas - commonly referred to as the "diagonal of low densities" - stretching from the northeast to the southwest, including alpines zones.

France geographical characteristics endow it with a diverse range of landscapes and natural environments, contributing to its ecological richness and tourism appeal. With 90 million international tourists in 2019, France is the world's leading tourist destination.

Territorial organization: France has a complex and structured territorial organization with several levels of local authorities. The organizational structure reflects both the geographical diversity of mainland France and its administrative history. It is part of a gradual decentralization process while maintaining a degree of centralization typical of the French state. Metropolitan France comprises 12 regions, 96 departments, and nearly 35,000 municipalities. Several authorities have special statuses, such as the City of Paris, the Metropolis of Lyon and the Collectivity of Corsica. Additionally, municipalities are grouped into Public Establishments for Inter-municipal Cooperation (EPCI), with 1,254 in 2024, including community groupings like "communauté de communes", "communautés d'agglomération" and metropolitan areas. EPCIs are legal entities formed by multiple municipalities that collaborate to deliver high-quality public services, either because they cannot sustain them individually or because coordination across large territory is more effective. They share responsibilities for mandatory functions (economic development, waste management, etc.) and optional ones (environmental protection, tourism development, social initiatives, etc.). Local authorities are not explicitly responsible for food, but they do take action in this area through other responsibilities, such as school catering (middle school for departments, high school for regions), social action or economic development.

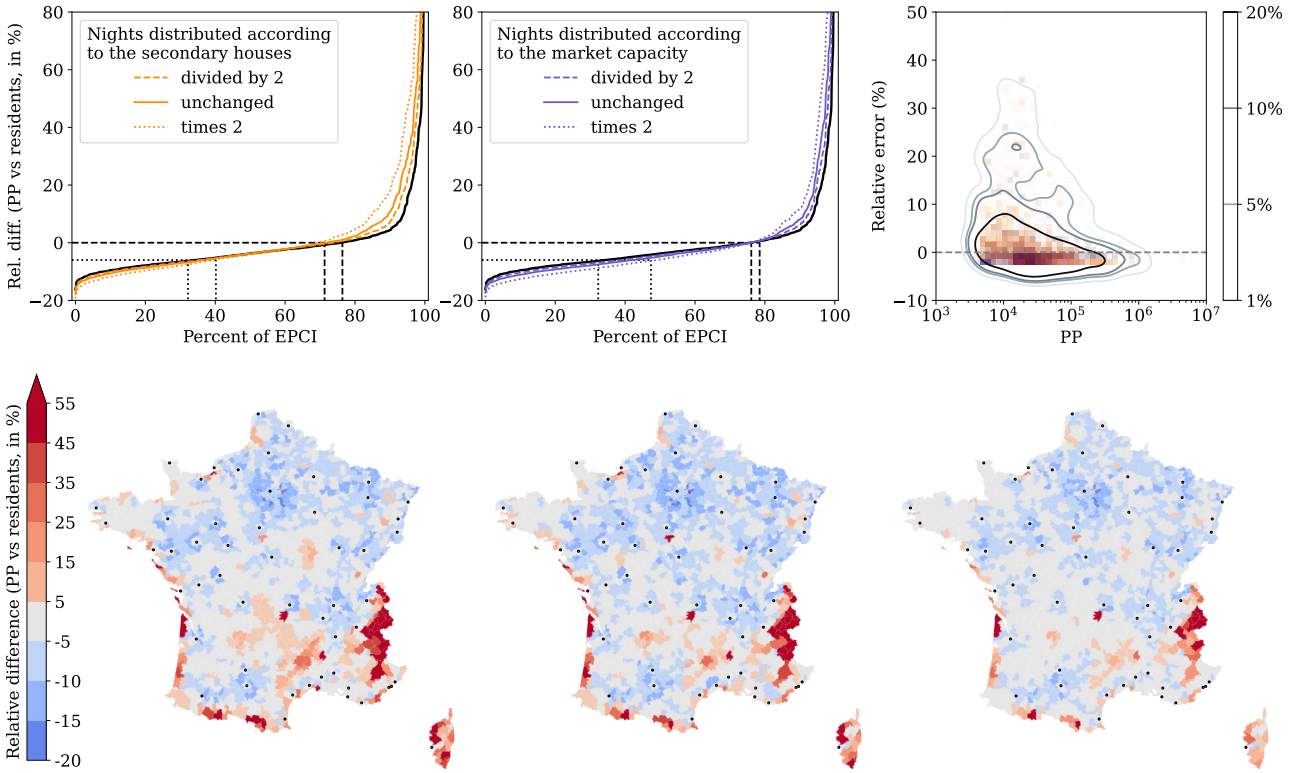


Figure A3: Sensitivity of the PP to the number of nights spent at friends and family and the way they are distributed spatially at the level of EPCIs. Left: impact of changing the distribution of overnight stays to follow that of secondary houses on the distribution of relative differences (top) and on the associated map (bottom). Center: distribution of overnight stays according to market capacity. The number of overnight stays is also varied: either divided by two (dashed lines) or multiplied by two (dotted lines). Bottom right: reference map for the distribution of overnight stay according to the residents. All maps represent the situation for the initial number of overnight stays. Top right : distribution of relative errors between the secondary houses or market capacity hypotheses and the reference hypothesis (residents) depending on the PP of the EPCI, for the original number of overnight stays.

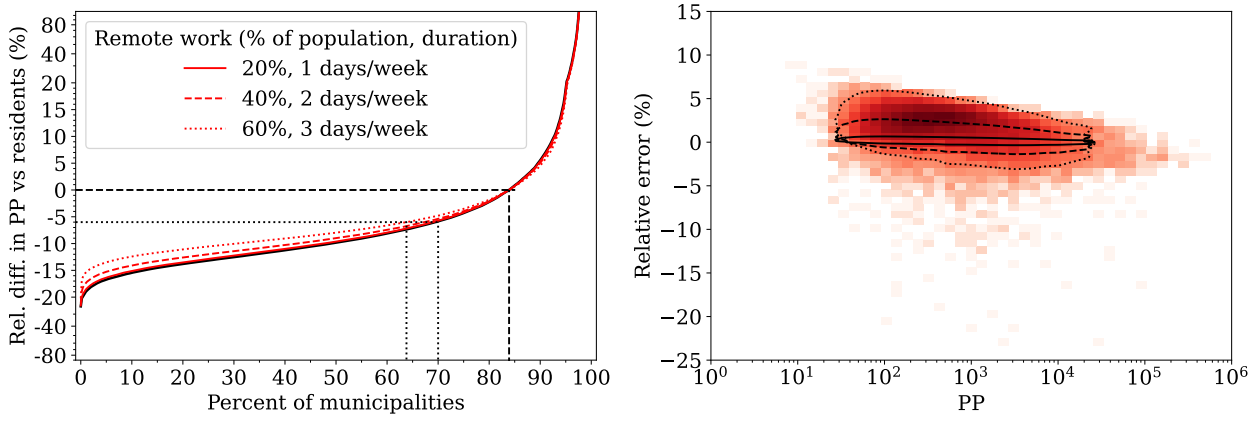


Figure A4: Impact of remote work on the PP under 3 different hypotheses. The first one is the current average for remote work in France: 20% of the population doing one day of remote work a week (continuous lines). The second is the average for the Île-de-France region, where Paris is located: 40% of the population working remotely for 2 days a week (dashed lines). Last hypothesis is the highest average rate of remote work for specific populations in that area, with 60% working remotely 3 days a week. On the left, the discrepancy that would arise in the relative difference between PP and residents is shown. On the right, the relative error between the data in the main text (not accounting for remote work) and the PP under the three remote work hypotheses is shown. The contour lines delimitate 95% of the distribution for each of the three hypotheses and the histogram show the full distribution for the third one, with the highest rate of remote work.